

Question Number	Answer	Additional guidance	Mark
5(a)(i)	an alternative form of a gene	accept version of a gene	(1)

Question Number	Answer	Mark
5(a)(ii)	The only correct answer is A ccddee <i>B is not correct because all alleles need to be recessive</i> <i>C is not correct because all alleles need to be recessive</i> <i>D is not correct because all alleles need to be recessive</i>	(1)

Question Number	Answer	Additional guidance	Mark
5(a)(iii)	A description that includes the following points: • polypeptide chain enters the rER (1) • {secondary/ tertiary} structure formed (1) • polypeptide is packaged into vesicles (by rER) / {(rER) vesicles fuse with / protein enters} Golgi (1) • carbohydrate added (to protein in Golgi) (1) • (glyco)protein is packaged into vesicles (by Golgi) and transported to cell surface membrane (1)	accept 3D structure formed / bonds form between different R groups accept transported in vesicle to the Golgi accept glycosylation occurs (in the Golgi) ignore modification occurs in Golgi without reference to carbohydrate being added accept (glyco)protein is packaged into vesicles (by Golgi) and vesicle fuses with cell surface membrane	(5)

Question Number	Answer	Additional guidance	Mark
5(b)	An explanation that includes three of the following points: - the greater the number of recessive alleles the lighter the colour of the seed / the greater the number of dominant alleles the deeper the red colour (1) - random assortment and crossing over results in (gamete) variation (1) - random fertilisation (of gametes) results in (seed) variation (1) - low probability of {inheriting 0/6 recessive alleles / egg cell with no recessive alleles being randomly fertilised by sperm cell with no recessive alleles} (1)	accept converse accept correct statement using information from table e.g. linking a colour to number of recessive alleles / linking colour to frequency / normal distribution accept the gametes vary in the number of {recessive/dominant} alleles they contain accept converse for chance of inheriting {3 recessive / 3 dominant} alleles	(3)